



Technical Service Bulletin

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CENSE™ New Rapid Rise Intake Manifold Temperature Fault Codes and Fault Code Latching

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

The purpose of This Technical Service Bulletin is to inform the field of a new CENSE™ fault code and to describe how certain fault codes will be latched 'active' when logged by the ECM.

A new CENSE™ diagnostic has been created to detect a rapid rise in intake manifold temperature. This feature can help detect intake valve failures. Four new fault codes are being activated for this diagnostic.

Rapid Rise Intake Manifold Temperature Fault Code

Fault Code	Description
783	Rapid rise in intake manifold temperature - Left bank, front quadrant
2157	Rapid rise in intake manifold temperature - Left bank, rear quadrant
2158	Rapid rise in intake manifold temperature - Right bank, front quadrant
2159	Rapid rise in intake manifold temperature - Right bank, rear quadrant

The new calibrations can be found on the September 2002 INCAL™ CD ROM for all QSK45 and QSK60 Industrial applications with the CENSE™ option.

To improve the engine protection feature, certain fault codes will be latched on once they become active. This means that once the fault code has been logged, the fault code will remain active until the fault code has been cleared using the INSITE™ CENSE™ electronic service tool. **Only** applications with the CENSE™ option will have this feature. Engines without the CENSE™ option will **not** have latched fault codes.

Fault Code Latching (Only the following fault codes will be latched)

Fault Code	Description	Effect on Engine*	Approximate Release Date
783	Rapid rise intake manifold temperature	Engine protection lamp lighted	September 2002

Fault Code Latching (Only the following fault codes will be latched)			
Fault Code	Description	Effect on Engine*	Approximate Release Date
2157	Rapid rise intake manifold temperature	Engine protection lamp lighted	September 2002
2158	Rapid rise intake manifold temperature	Engine protection lamp lighted	September 2002
2159	Rapid rise intake manifold temperature	Engine protection lamp lighted	September 2002
612	Oil filter restriction	Non engine protection lamp lighted	September 2002
555	High blowby pressure	Engine derate	September 2002
143	Low oil pressure	Engine derate	September 2002
415	Very low oil pressure	Engine derate	September 2002

* If the application is an electric-drive haul truck, then the truck will be kicked out of propel as well as an engine derate when the fault is active.

Note : There is a software issue with the June/July/August CENSE™ Calibration CD. Do not use the June/July/August CD to recalibrate CENSE™ ECMs.

This issue is due to a software bug. This results in a derate occurring when intake manifold temperature is greater than 77° C [170° F] although no fault code is logged. The software issue has been resolved on the September CENSE™ Calibration CD.

Description of Symptoms

- Operator claims engine is **only** running at 1675 rpm or less
- Technician can **not** find a fault code which causes derate
- Trouble occurs during high ambient temperatures.